

Case Study Report – AntiGravity Implementation

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Course / Subject: Android development using Gen AI

Tool Used: AntiGravity

Laptop: ASUS TUF Gaming A15

Operating System: Windows 11

RAM: 8 GB

GPU: NVIDIA GeForce GTX 1650 Ti

1. Introduction

This report presents the execution of a simple case study using the AntiGravity tool. The objective of this activity was to understand how AntiGravity can be used to explore ideas, test simple implementations, and support the development workflow. The case study focuses on solving a practical problem and documenting the entire process from problem definition to final output.

2. Problem Definition

Students often struggle to manage their daily study tasks and assignments. Without a proper tracking system, tasks may be forgotten or delayed, which affects productivity and academic performance.

The problem addressed in this case study is to create a **simple Student Study Tracker** that helps users:

- Add study tasks
- Mark tasks as completed
- View all tasks and their progress

The aim is to develop a simple system that improves task organization and time management for students.

3. Solution Design

To solve the problem, a basic system was designed with the following functionality:

1. **Task Creation** – Users can add new study tasks.
2. **Task Status Update** – Users can mark tasks as completed.
3. **Task Display** – The system shows a list of tasks with their completion status.

The design focuses on simplicity so that users can easily track their daily academic activities.

4. Implementation Using AntiGravity

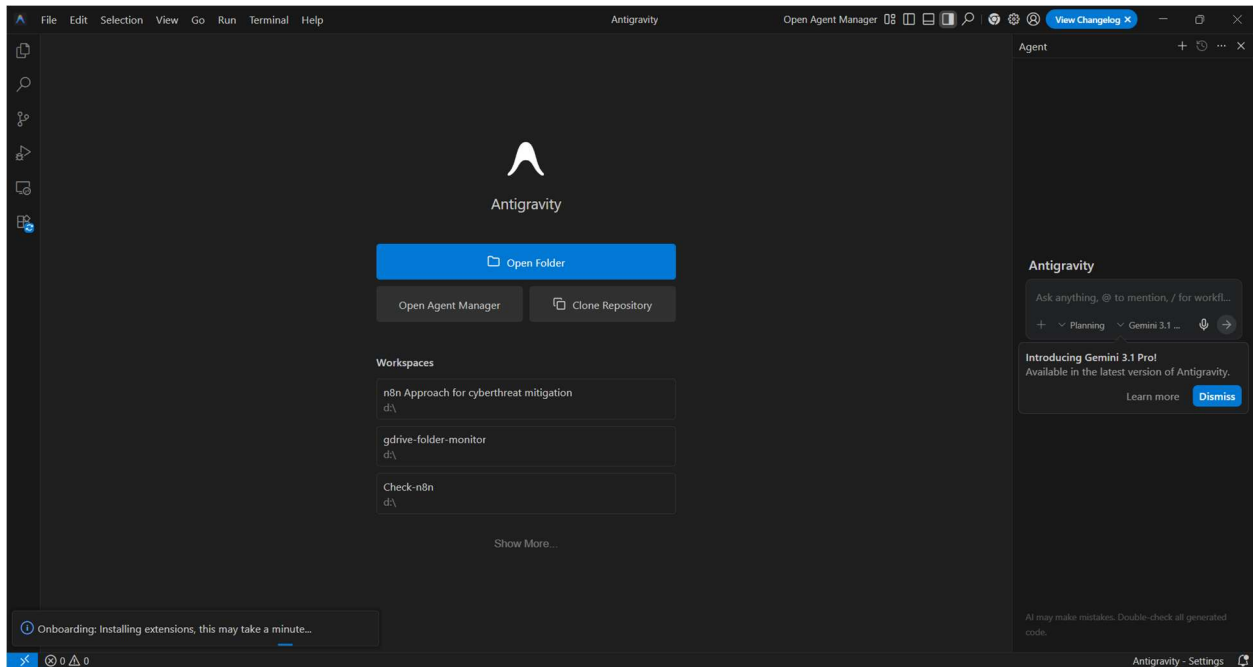
AntiGravity was used during the development process to experiment with the logic and structure of the solution.

The implementation process included the following steps:

1. Defining the problem and planning the logic of the study tracker.
2. Creating a simple structure to store study tasks.
3. Implementing functions to add tasks and update their status.
4. Displaying the task list along with completion information.
5. Testing the logic using AntiGravity to ensure the system works correctly.

AntiGravity helped simulate the behavior of the program and allowed quick testing of ideas during development.

Screenshot 1: AntiGravity interface / setup



Screenshot 2: Code or logic implementation

```
// Data class to represent a study task
data class StudyTask(
    val name: String,
    var completed: Boolean = false )
// Function to display all tasks
fun displayTasks(tasks: List<StudyTask>) {
    println("\nStudy Task List:")
    tasks.forEachIndexed { index, task ->
        val status = if (task.completed) "Completed" else "Pending"
        println("${index + 1}. ${task.name} - $status")    }    }
fun main() {
    val tasks = mutableListOf<StudyTask>()
    // Adding tasks
    tasks.add(StudyTask("Complete Kotlin Assignment"))
    tasks.add(StudyTask("Revise Android Studio Concepts"))
    tasks.add(StudyTask("Prepare for Programming Test"))
    // Display tasks before completion
    println("Before completing tasks:")
    displayTasks(tasks)
    // Mark a task as completed
    tasks[0].completed = true
    // Display tasks after completion
    println("\nAfter completing a task:")
    displayTasks(tasks) }
```

5. System Output

After implementing the solution, the system successfully performed the following operations:

- Added new study tasks to the list
- Displayed all tasks
- Updated task status when marked as completed

The output clearly showed the list of tasks along with their completion status, helping users monitor their study progress.

Screenshot 3: Program output showing tasks

```
Before completing tasks:
```

```
Study Task List:
```

- ```
1. Complete Kotlin Assignment - Pending
2. Revise Android Studio Concepts - Pending
3. Prepare for Programming Test - Pending
```

```
After completing a task:
```

```
Study Task List:
```

- ```
1. Complete Kotlin Assignment - Completed  
2. Revise Android Studio Concepts - Pending  
3. Prepare for Programming Test - Pending
```

6. Analysis

The case study demonstrates that even a simple task tracking system can improve productivity and organization. AntiGravity made it easier to test different approaches and modify the logic quickly.

The tool supported experimentation and allowed rapid testing without needing a full development environment.

7. Possible Improvements

Although the current solution is simple, it can be extended with additional features such as:

- Adding deadlines for tasks
- Setting reminders or notifications
- Saving tasks permanently using a database
- Creating a graphical user interface (GUI)
- Tracking weekly or monthly progress

These improvements would make the system more useful and scalable.

8. Conclusion

This case study demonstrated how AntiGravity can assist in solving practical problems through experimentation and rapid development. The Student Study Tracker successfully addressed the problem of managing study tasks.

Through this activity, I gained a better understanding of problem-solving, system design, and the importance of testing ideas during development. AntiGravity proved to be a helpful tool for exploring solutions and improving the development workflow.
